Lesson Plan · Thursday, November 5

Algebra I · Mr. Pandey · Module 3, Topic 1, Lesson 1: Solving Linear Equations · 150-day pacing: 1 day · essentials only (✪)

Objectives (students will…)

- Write equivalent equations using the properties of equality.
- Solve linear equations (distributive property, variables on both sides) and name the property for each step.
- Tell whether an equation has one solution, no solution, or infinite solutions.

TEKS

- **A.5A** (readiness) solve linear equations in one variable, including distributive property and variables on both sides
- Process: A.1D, A.1F, A.1G

Key terms

- solution · no solution · infinite solutions

Materials

- Student Edition pp. 345–350 (Getting Started, Activities 1.1, 1.2)
- Properties of equality + number properties anchor chart (SE pp. 347–348)
- Whiteboards, markers; algebra tiles for Just-in-Time support
- Exit ticket (4 questions below) on half sheets

Bell work (on the board)

- $3x + 5 = 20$ $x = 5$
- $2(x - 4) = 10$ $x = 9$
- Simplify $4x + 3 - x + 2$ $3x + 5$
- Is $x = 2$ a solution of $5x - 1 = 9$? yes, $9 = 9$

Agenda (60 minutes)

Time	Activity	What to do / say
0–5	Bell work	4 warm-up items above. Go over Q2 (distribute first) and Q4 (substitution checks a solution) — both are used today.
5–20	✪ Getting Started: Equation Creation · SE p. 346	Read the definition of *solution* together. Pairs start from $x = 2$ and do Q1 a–d (add a constant, multiply, subtract ax, divide), e.g. $x + 3 = 5 \rightarrow 4x + 12 = 20 \rightarrow 2x + 12 = -2x + 20 \rightarrow x + 6 = 10 - x$. Q2: swap with a partner and check $x = 2$ by substitution. **Look for:** multiplying/dividing only the x-term instead of *every* term.
20–40	✪ Activity 1.1: Using Properties to Justify Solutions · SE pp. 347–348	Explicit teach: properties of equality table (add, subtract, multiply, divide). Q1: verify Harper ($-3x + 14 = 18 - 5x$) and Sofia ($2x = 5 - \frac{1}{2}x$) both give $x = 2$ — substitution $8 = 8$, $4 = 4$, then properties. Read number properties (commutative, associative, distributive). Q2 pairs, name each property: (b) $0.6(2x + 1) = 4x + 6.2$ $x = -2$; (c) $\frac{1}{2}x - 6 = 2 + (2x + 1)$ $x = -6$. Q3 Daniel vs. Chloe, $4(3x + 2) = 8x + 4$ $x = -1$: Daniel distributes first, Chloe divides by 4 first — more than one correct path.
40–50	✪ Activity 1.2: Solutions to Linear Equations · SE pp. 349–350	Q1: $2 = 2$ stays true no matter what you do to both sides (infinite solutions). Q2: $2 = 3$ stays false (no solution). Read definitions. Q4: $2x = 3x$ one solution, $x = 0$. Board rule (simplify each side first): same x-coefficient, same constant → **infinite** · same x-coefficient, different constant → **none** · different x-coefficients → **one**.
50–58	Exit ticket (independent)	1. Solve $3(x - 2) = x + 8$ and name each property. distributive → $3x - 6 = x + 8$; subtraction → $2x - 6 = 8$; addition → $2x = 14$; division → $x = 7$ 2. $2(x + 3) = 2x + 6$ infinite 3. $4x - 1 = 4x + 5$ no solution 4. $5x + 2 = 3x + 2$ one, $x = 0$ IEP/504: read aloud, properties chart allowed, extra time.
58–60	Wrap up	Students answer the Essential Question (SE p. 345). Homework: Lesson 1 Assignment, SE pp. 355–356 (Practice 1a–c, 2a–f). Next class Mon 11/9: Lesson 2, Literal Equations.

Check for understanding

- Whiteboards: "What property lets you add 4x to both sides?" addition property of equality
- Thumbs: "$x + 5 = x + 5$ has no solution." Down — infinite
- Activity 1.1 Q2–3 and 1.2 Q3 are the TE's assessment points.

After class (exit ticket)

- Missed Q1 → Skills Practice Set A (solving + justifying).
- Missed Q2–4 → Skills Practice Set B (number of solutions).
- All correct → Skills Practice Set C (variables on both sides).

Skipped (not starred, district gives Lesson 1 one day): Activity 1.3 Tic-Tac-Bingo (35–40 min) and Talk the Talk (SE p. 352). Tic-Tac-Bingo fits a Learning Individually day if time opens up. TE plans 2 days; this is TE Day 1.